

ICMP Guide: Wireshark Analysis of ICMP Traffic

What is ICMP?

ICMP (Internet Control Message Protocol) is a supporting protocol used for error reporting, diagnostics, and network troubleshooting. It is often described as the "messaging" protocol of IP networks.

Unlike TCP or UDP, ICMP is not used for transferring user data — it helps devices communicate about network conditions.

Common Types & Codes

The most important fields in ICMP are **Type** and **Code**:

Type	Name	Common Use	Code
0	Echo Reply	Ping response	0
8	Echo Request	Ping request	0
3	Destination Unreachable	Host/network unreachable	0-15
11	Time Exceeded	TTL expired (traceroute)	0
5	Redirect	Better route suggested	-

Typical Use Cases

- **Ping** (Type 8 → Type 0)
- **Traceroute** (uses Time Exceeded messages)
- Error reporting (port unreachable, fragmentation needed, etc.)

Why Analysts Care About ICMP

- Reconnaissance: Attackers use ICMP for host discovery and network mapping.
- Tunneling & Exfiltration: Data can be hidden inside ICMP packets.
- Diagnostics: Helps understand why connections fail.
- DoS attacks: ICMP floods, ping of death, etc.
- Mixed-traffic forensics: Often appears alongside other protocols.

Wireshark Fundamentals for ICMP Analysis

1. Basic Filtering

Bash

```
icmp # All ICMP traffic
```

```
icmp.type == 8      # Echo Request (Ping)
icmp.type == 0      # Echo Reply
icmp.type == 3      # Destination Unreachable
icmp.type == 11     # Time Exceeded (Traceroute)
icmp.contains "flag" # Search for text inside payload
icmp.data.size > 100 # Large ICMP packets (possible tunneling)
```

2. Key Fields to Examine

- **Type and Code**
- **Identifier and Sequence Number** (used to match requests/replies)
- **Data / Payload** — Can contain arbitrary data (including hidden flags)
- **Checksum**
- **Time to Live (TTL)** in the IP header of ICMP packets

Common ICMP Analysis Scenarios in Thaghrach Challenges

1. **Basic Ping Analysis**
 - Measure round-trip time and packet size.
 - Detect live hosts.
2. **Data Hidden in ICMP**
 - Flags or messages embedded in the data portion of Echo packets.
3. **Traceroute Reconstruction**
 - Follow increasing TTLs and Time Exceeded responses to map the path.
4. **ICMP Tunneling**
 - Large or unusual payloads in Echo Request/Reply.
5. **Mixed Traffic Forensics**
 - ICMP combined with TCP/HTTP/DNS to understand full attack scenarios.
6. **Error Message Analysis**
 - Why a connection failed (Destination Unreachable).

Practical Workflow in Wireshark

1. Apply icmp filter.
2. Expand the **Internet Control Message Protocol** section.
3. Look at **Type** and **Code** first.
4. For Echo packets, examine the **Data** field — this is where flags are often hidden.
5. Use **Statistics** → **ICMP** for summary of types and counts.
6. **Follow** → **IP Stream** or **TCP Stream** when ICMP appears with other protocols.

Pro Tip: Right-click an ICMP packet → **Follow** → **ICMP Stream** to see the full conversation.

Security Red Flags in ICMP Traffic

- Large ICMP packets (possible data exfiltration or tunneling).
- High volume of Echo Requests (ICMP flood).
- ICMP packets with suspicious or encoded data.
- Unexpected Destination Unreachable messages (possible reconnaissance).

- ICMP used from external IPs to internal networks.

Example Challenge Goals You Will Encounter

- Extract the hidden flag from the data portion of ICMP Echo packets.
 - Reconstruct a message sent piece-by-piece through multiple ICMP packets.
 - Perform a virtual traceroute from the capture.
 - Analyze why a TCP connection failed using ICMP error messages.
 - Identify signs of ICMP-based covert communication.
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Tips for Thaghras Students:

- ICMP is simple but powerful for hiding information.
- Always check the **Data** field in Echo Request/Reply packets.
- Combine ICMP analysis with other protocols in the same capture.
- Pay attention to **Sequence Numbers** — they help match requests and replies.